



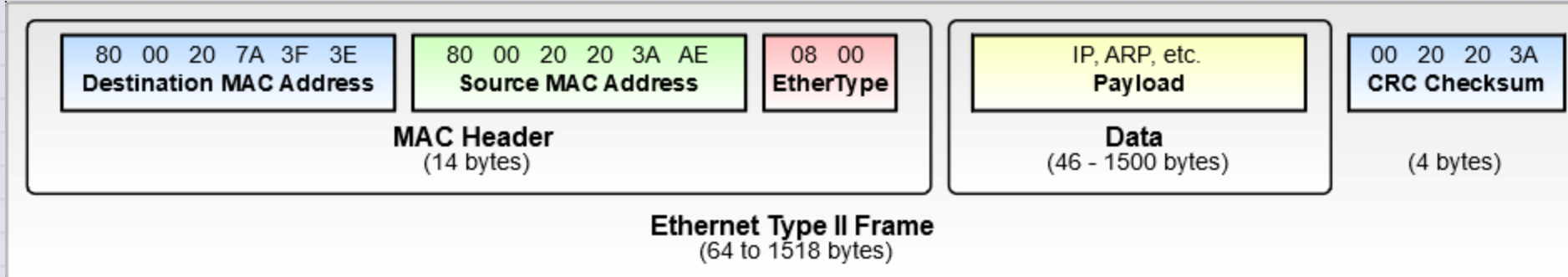
**ACIT 2620**

# Principles of Enterprise Networking

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# Ethernet

# Frames



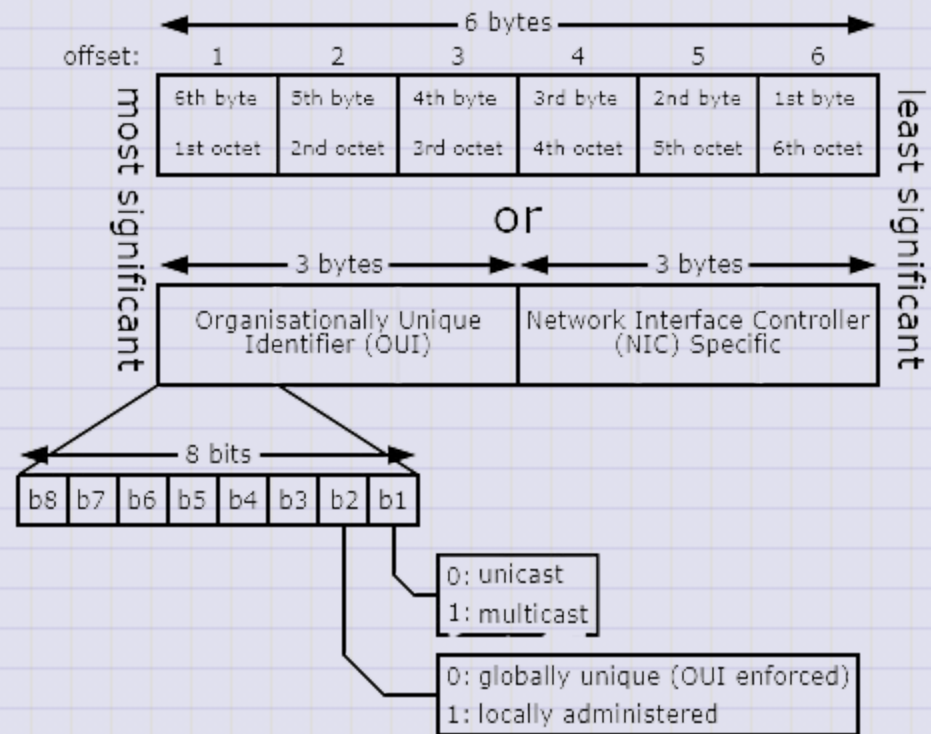
- Preamble
  - Marks beginning of entire frame
- Start of Frame Delimiter (SFD)
  - Indicates beginning of addressing fields
- Destination Address
  - Contains destination node address

- Source Address
  - contains address of sender node
- Length (LEN)
  - indicates length of data/payload
- Data (payload)
  - contains data, or segmented part of data, transmitted from originating node

- Pad
  - Used to increase size of the frame to its minimum size requirement of 46 bytes
- Frame Check Sequence
  - provides algorithm to determine whether data were correctly received
  - most commonly used algorithm is Cyclic Redundancy Check (CRC)

# Ethernet Addressing

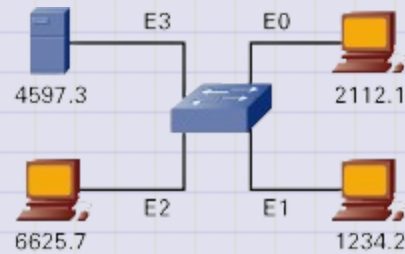
- MAC address: Media Access Control (MAC) sub-layer
- 48 Bits
- Number uniquely defining a network node
- Generally rendered as Hex: **00:1e:33:ba:87:c1**
- Doesn't contain any data regarding network location –just an ID



- first three bytes
  - Either Manufacturer hard coded
  - Or Reserved Addresses (common ones)
    - Broadcast Address **FF:FF:FF:FF:FF:FF**
    - Spanning Tree Multicast: **01:80:C2:00:00:00**
    - IANA reserves all address starting with **00:00:5E** see Ethernet Numbers(this includes IPv4 multicast -and inserts the low 23 Bits of the multicast IPv4 Address into the Ethernet Address)
    - **33:33:XX** is reserved for IPv6 Multicast

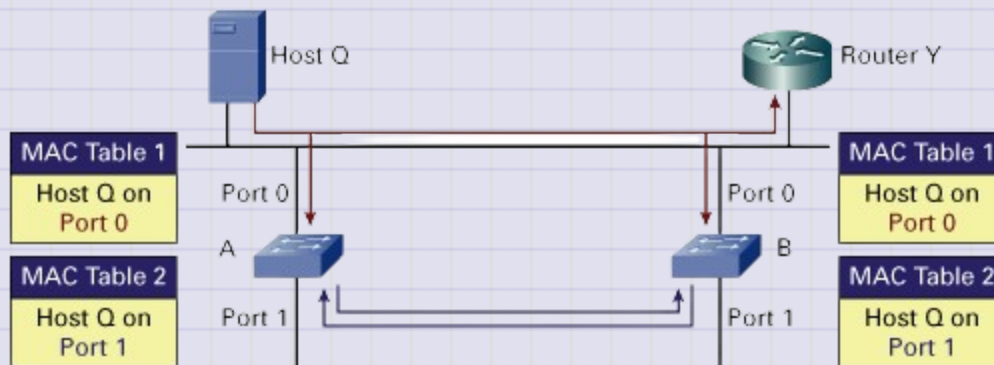
# Switching

- Making Forwarding decisions
  - Transparent bridging

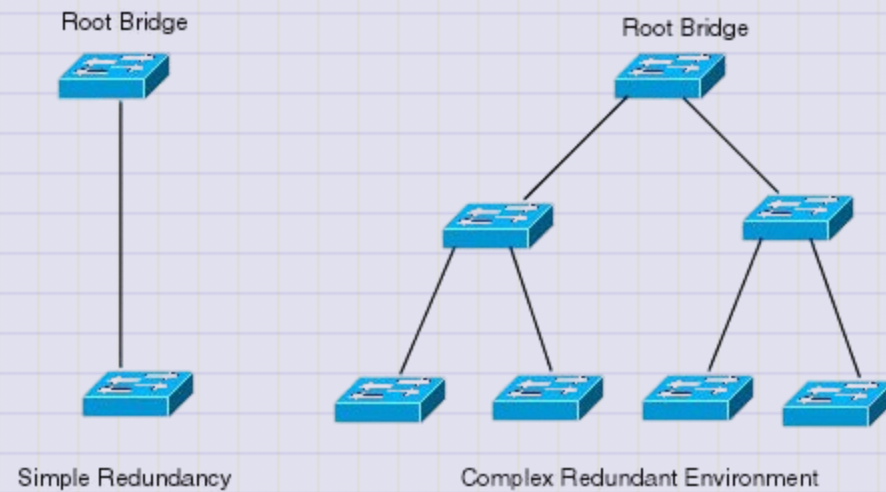


| Time 0 |         | Time 1 |         | Time 2 |         |
|--------|---------|--------|---------|--------|---------|
| Port   | Address | Port   | Address | Port   | Address |
|        |         | E2     | 6625.7  | E2     | 6625.7  |
|        |         |        |         | E0     | 2112.1  |

# Broadcast Loop and STP

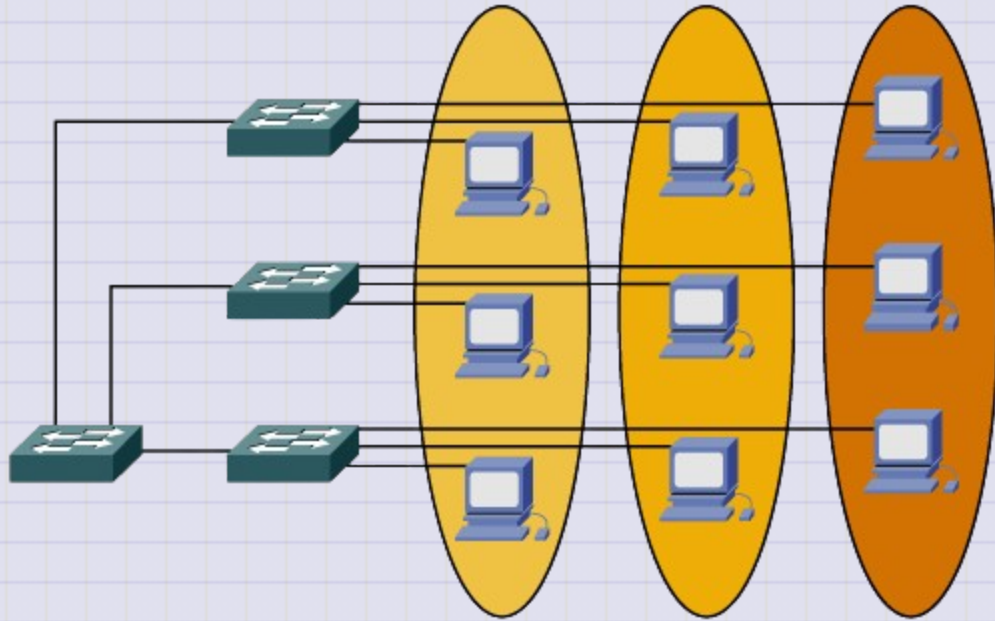


# Spanning Tree Protocol



# VLANs

- Virtual Local Area Networks
- A logical network within a physical network
- Achieved by grouping some of the switch ethernet ports into a logical broadcast domain
- Can span multiple switches

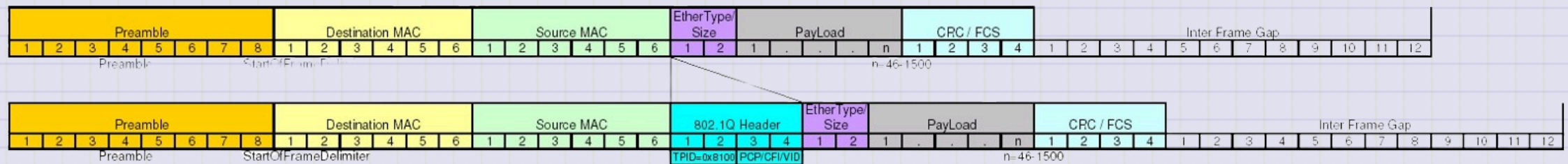


## VLAN port types

- Access ports
  - assigned VLAN ID
  - for connecting end hosts/nodes
  - nodes connected to ports with same VLAN ID are in the same broadcast domain

- Trunk ports
  - typically for switch to switch or switch to router connection
  - carry "tagged" frames, i.e modified ethernet frames with VLAN markers

# Tagged frames



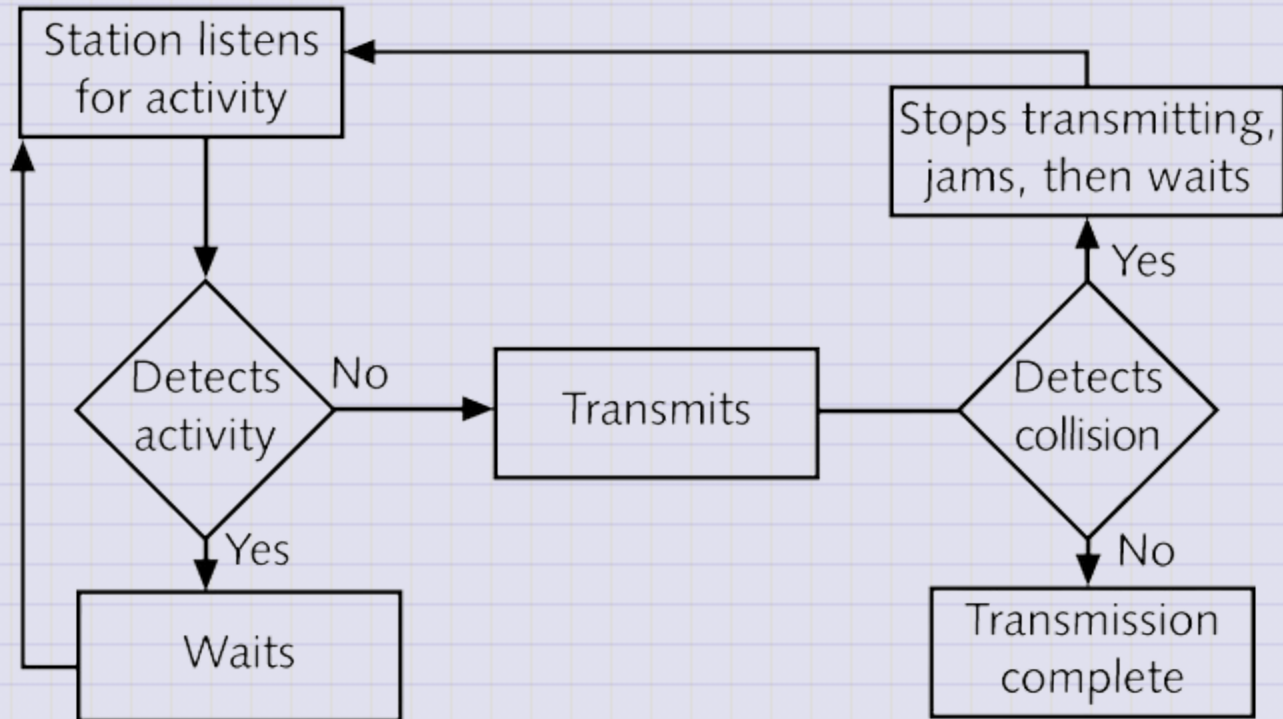
- 4-byte tag header inserted between Source MAC and EtherType fields
  - 2-byte tag protocol identifier (TPID)
    - a fixed value of 0x8100 that indicates the frame carries tag information.

- 2-byte tag control information (TCI)
  - Three-bit user priority (used to prioritize traffic)
  - Drop Eligible Indicator (DEI) (in congestion is frame “dropable”)
  - Twelve-bit VLAN identifier (VID)-Uniquely identifies the VLAN to which the frame belongs

# Link Access Methods

- Manage shared medium access contention (collision)
- Two methods:
  - CSMA/CD: for wired Ethernet
  - CSMA/CA: for wireless Ethernet

## CSMA/CD



# CSMA/CA

